SIGMACAMP · ELECTROCHEMISTRY LAB · SESSION 2

Copper Electroplating

Lecture & Instructor Notes

Audience: instructors and teaching assistants

Purpose: Teaching script, deep-dive notes, and coaching guidance

Session hook: Students visibly transform a metal object

Length: 90-minute lab block (~20 min concepts + ~70 min hands-on)

This packet includes

1. Session overview & plan
2. Lecture talking points
3. Instructor deep-dive notes
4. Preparation reminders

Generated from the Electrochemistry workshop curriculum. Prefer the latest PDF from the course repository if procedures change mid-week.

SESSION OVERVIEW & PLAN

Session 2 — Copper Electroplating

Hook: Students visibly transform a metal object.

Learning outcomes

- Explain electrolysis and distinguish it from a galvanic cell
- Identify cathode and anode in an **electrolytic** cell
- Describe how current controls deposition rate
- Explain why surface preparation and current density affect deposit quality

Session plan (90 min)

Time	Activity	Reference
0–10 min	Show blue $CuSO_4$ solution / dissolving crystals; preview plating	lecture.md — Hook
10–25 min	Concept: ion deposition at cathode, anode dissolution	lecture.md
25–35 min	Clean cathode object (sand, rinse, no fingerprints)	experiment.md — Prep
35–50 min	Set up plating cell	experiment.md — Setup
50–70 min	Plate 2–10 min; observe every 1–2 min	experiment.md — Plating
70–80 min	Compare variables	experiment.md — Variables
80–90 min	Discussion: surface prep and current density	lecture.md — Wrap-up

Key reactions

```
Cathode (−): Cu²⁺(aq) + 2e⁻ → Cu(s)
Anode (+):   Cu(s) → Cu²⁺(aq) + 2e⁻
```

Files

- lecture.md — Concept block
- experiment.md — Plating procedure
- materials.md — Checklist
- preparation.md — $CuSO_4$ solution prep

Status

- [ ] 0.5 M $CuSO_4$ mixed and labeled
- [ ] Plating procedure pilot-tested
- [ ] Lecture slides/diagrams ready

LECTURE TALKING POINTS

Session 2 — Lecture: Electrolysis and Copper Plating

Target duration: ~20 minutes

> **Instructors:** galvanic vs electrolytic signs, Cu^{2+} chemistry, current density — instructor.md.

Opening hook (0–10 min)

Demo / inspection

1. Hold up blue **$CuSO_4$** crystals and/or the pre-mixed blue plating bath
2. Ask: "*Why is this liquid blue? What dissolved particles make plating possible?*"
3. Optional: dissolve a pinch of $CuSO_4$ in water in front of the class

Guiding questions

- Is this the same as the lemon battery, or the reverse?
- Who is supplying the energy now?

Core concepts (10–25 min)

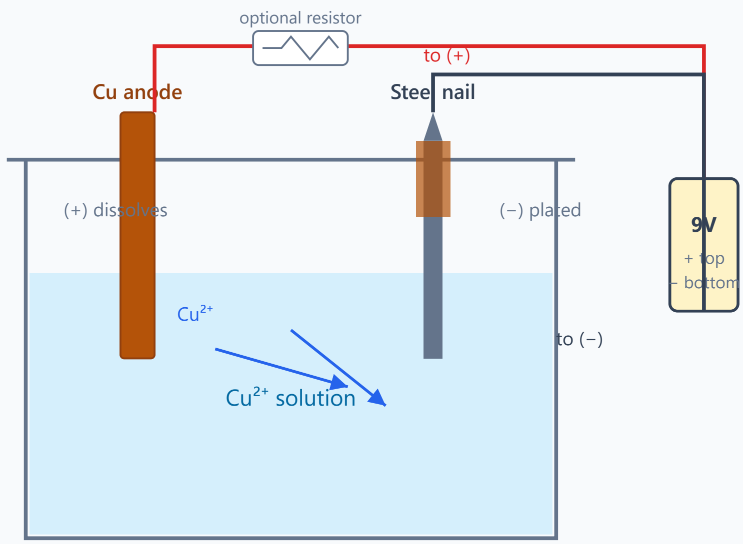
1. Galvanic vs. electrolytic cells

	Galvanic (Session 1)	Electrolytic (Session 2)
Energy	Chemical → electrical	Electrical → chemical
Spontaneous?	Yes	No — needs power supply
Anode sign	Negative (−)	Positive (+)
Example	Fruit battery	Copper plating

2. What happens in the plating bath

- **Cathode (−):** Object to plate — Cu^{2+} ions gain electrons → **Cu metal deposits**
- **Anode (+):** Copper metal — Cu dissolves → **replenishes Cu^{2+} in solution**
- Ions carry charge through the solution; electrons through the wire and power supply

Session 2 - Copper electroplating (electrolytic cell)



Cathode (−): $Cu^{2+} + 2e^- \rightarrow Cu(s)$ - copper deposits on nail

Anode (+): $Cu(s) \rightarrow Cu^{2+} + 2e^-$ - copper dissolves, replenishes bath

Power supply drives non-spontaneous reaction (opposite of Session 1 fruit battery)

3. Current controls deposition rate

- More current → faster plating (more electrons per second → more Cu atoms)
- **But:** too much current → rough, dark, powdery deposit (bad current density)

4. Smooth vs. rough deposits

Factor	Good deposit	Bad deposit
Surface	Clean, sanded	Greasy, oxidized
Current	Moderate, steady	High, uncontrolled
Distance	Electrodes reasonably spaced	Too close → hot spots
Time	Enough for even layer	Too fast → tree-like growth

5. Why cleaning matters

- Oil from fingers blocks electron transfer at the surface
- Oxide layers prevent adhesion
- **Rule:** Sand → rinse → handle by edges only → plate immediately

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Wrap-up discussion (80–90 min)

- Why must the object connect to the **negative** terminal?
- What would happen with a steel anode instead of copper?

- How is this related to manufacturing (chrome on bumpers, gold on jewelry)?

Visual aids to prepare

- [ ] Side-by-side: fruit battery vs. plating cell (energy direction arrows)
- [ ] Photo series: 0 min, 2 min, 5 min plating on nail
- [ ] Examples of rough vs. smooth copper (if available from pilot)

Vocabulary

- [ ] Electrolysis
- [ ] Electrolyte / electrolytic cell
- [ ] Cathodic deposition
- [ ] Anodic dissolution
- [ ] Current density

Bridge to Session 3

"We moved copper atoms today without counting them. Next session we plate silver and use Faraday's law to calculate exactly how many atoms we deposited — and how thick the film is."

INSTRUCTOR DEEP-DIVE NOTES

Session 2 — Instructor Notes: Copper Electroplating Deep Dive

Audience: instructors and TAs only. Pair with lecture.md and experiment.md.

Why this session exists

Session 1 made electricity from chemistry. Session 2 **forces** chemistry with electricity: copper ions become solid metal on a student's object. Visually dramatic, conceptually the reverse of a galvanic cell. Students must leave knowing: **cathode = reduction = object being plated**, and the power supply — not spontaneous chemistry — drives the process.

Atomic / ionic picture of plating

What Cu^{2+} is

Copper metal atoms have valence electrons that can be removed:

```
Cu(s) → Cu²⁺(aq) + 2e⁻   (oxidation / dissolution)
Cu²⁺(aq) + 2e⁻ → Cu(s)   (reduction / plating)
```

Cu^{2+} in water is typically a hydrated complex (often written $Cu^{2+}(aq)$ or $[Cu(H_2O)_6]^{2+}$). The blue color of many copper solutions comes from these hydrated ions (and related species). Students only need: *blue liquid ≈ dissolved copper ions ready to plate.*

Why two electrons matter

Each Cu^{2+} needs **two** electrons to become one Cu atom. Faraday's law (Session 3) makes this quantitative; here it explains rate qualitatively:

- Higher current → more electrons per second → more Cu atoms deposited per second
- Too high current density → ions cannot arrive / arrange orderly → powdery, dark deposits

Valence and periodic context

Element	Common ion in this lab	Notes
Cu	Cu^{2+}	Transition metal; d-electrons give colored solutions
Fe (nail)	Fe^{2+} / Fe^{3+} possible side products	Steel cathode is a **substrate**, not the ion being plated
H	H^+	Competing reduction → H_2 bubbles if overpotential / acidity allows

Copper sits **below** hydrogen and iron in the activity series for many contexts — which is why Cu^{2+} is relatively easy to reduce onto steel. That is also why iron nails rust when left wet, but copper plating can stick if the surface is clean.

Galvanic vs electrolytic — master the signs

Feature	Session 1 galvanic	Session 2 electrolytic plating
Energy direction	Chemical → electrical	Electrical → chemical
Spontaneous?	Yes	No
Anode	Zn oxidizes (− terminal)	Cu oxidizes (+ of battery)
Cathode	H^+ reduced (+ terminal)	Cu^{2+} reduced (− of battery)
Who supplies energy?	Chemical reactants	Battery / supply

Classroom one-liner: *"Yesterday zinc wanted to give away electrons. Today we force copper ions to take electrons by connecting the object to the negative terminal."*

Half-reactions in the plating bath

Preferred teaching set (copper anode)

Cathode (−) — object:

```
Cu²⁺(aq) + 2e⁻ → Cu(s)
```

Anode (+) — copper metal:

```
Cu(s) → Cu²⁺(aq) + 2e⁻
```

Net: copper is transferred from anode to cathode. Solution Cu^{2+} concentration stays roughly constant if the anode is copper (idealized).

If the anode is *not* copper (graphite, steel)

Anode reaction may become water oxidation or anode metal dissolution:

```
2H₂O → O₂ + 4H⁺ + 4e⁻   (water oxidation, simplified)
```

Then Cu^{2+} in solution is **consumed** and not replenished — bath depletes, plating slows, quality suffers. Emphasize why a copper anode is preferred.

Competing cathode reaction

```
2H⁺ + 2e⁻ → H₂(g)
```

Bubbles on the object mean some current is wasted on hydrogen instead of copper. Acidic baths and high current density increase this. Dark/powdery deposits often correlate with high current and side reactions.

Electrochemical series connection

Approximate E° values:

Couple	E° (V)
Cu^{2+}/Cu	+0.34
H^+/H_2	0.00
Fe^{2+}/Fe	−0.44
Zn^{2+}/Zn	−0.76

Plating Cu onto Fe is thermodynamically reasonable: Cu^{2+} is a stronger oxidant than Fe^{2+} (Cu^{2+} more easily reduced). Students do not need the calculation; you can say: *"Copper ions are 'eager' to become metal compared with iron ions — so copper coats the nail when we supply electrons."*

Current density and deposit quality

Current density ≈ current / cathode area.

Too low	Just right	Too high
Very slow, thin coat	Smooth, adherent, coppery	Burnt, black, powdery, treeing

Instructor controls if groups struggle:

- Series resistor (100–500 Ω) with 9 V pack
- Larger anode–cathode distance
- Shorter plating times (2–5 min first)
- Cleaner surface prep

Why cleaning matters (surface chemistry)

Oils and oxides:

- Block electron transfer
- Prevent metallic bonding of new Cu atoms to the substrate
- Cause peeling or patchy color

Protocol: sand → rinse → handle by edges → plate immediately. Fingerprints are a real failure mode.

Experiment coaching notes

$CuSO_4$ plating bath

- Use **0.5 M $CuSO_4$** mixed before class (tap water OK).
- Blue color = hydrated Cu^{2+} ready to plate — a useful visual for students.
- Acceptable improvised range: ~0.25–1.0 M if you must adjust.

- **Waste:** collect copper solutions; do not pour casually down drains.

Wiring checklist (walk the room)

1. Object on **battery negative (−)**
2. Copper strip/wire on **battery positive (+)**
3. Electrodes not touching
4. Both immersed in solution
5. Optional ammeter in series

Students reverse polarity constantly. A reversed cell can etch the object or plate onto the “wrong” electrode.

What good vs bad looks like

Appearance	Interpretation
Bright salmon/copper color	Good reduction, moderate current
Dark brown / black powder	Too much current; wipe/stop/reduce I
Patchy	Dirty surface or poor immersion
Bubbling heavily on object	Competing H_2 evolution

Variable tests (how to interpret group data)

- **Time:** thicker coat, then roughness if too long
- **Distance:** closer → higher current → faster but riskier
- **Resistor:** lower current → slower, often prettier
- **Surface prep:** dirty control looks obviously worse — best demo variable

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Misconceptions to preempt

Misconception	Correction
“The battery stores the copper.”	Battery supplies **electrons**; copper comes from ions / anode.
“Positive terminal plates the object.”	Object must be **negative** (cathode).
“Color in solution is paint.”	Color is from **dissolved Cu^{2+} ions**.
“More voltage always better.”	Excess driving force → burnt deposits.

Board plan

1. Draw Session 1 cell vs Session 2 cell side by side (energy arrows opposite).
2. Label anode/cathode signs for **electrolytic** case.
3. Write both half-reactions; circle “object = cathode.”
4. Show photo timeline: 0 / 2 / 5 min plating.
5. Bridge: next time we **count** atoms with Faraday’s law using silver.

Optional enrichment

- Overpotential and why real plating baths use additives (levelers, brighteners).
- Hull cell concept for current-density mapping (advanced).
- Industrial electroplating: chrome, nickel, PCB copper deposition.

PREPARATION REMINDERS

Session 2 — Preparation

Before Session 2 (instructor)

- [ ] Prepare **0.5 M $CuSO_4$** plating bath (e.g. 62 g $CuSO_4 \cdot 5H_2O$ per 500 mL tap water); label concentration
- [ ] Pilot plate one nail: record time, current, appearance
- [ ] Pre-sand a few demo objects
- [ ] Source resistors if using current limiting

Mixing 0.5 M $CuSO_4$ (quick recipe)

Batch size	$CuSO_4 \cdot 5H_2O$ (blue crystals)	Water
200 mL	**25 g**	Fill to 200 mL
500 mL	**62 g**	Fill to 500 mL
1.0 L	**125 g**	Fill to 1.0 L

1. Weigh $CuSO_4 \cdot 5H_2O$; dissolve in ~¾ of the water; stir until clear blue.
2. Top up to final volume; label **"0.5 M $CuSO_4$ — Session 2 — do not drink"**.
3. Tap water is fine; use distilled only if hard tap water makes the bath milky.

Day-of setup

- [ ] Goggles and gloves at each station
- [ ] Aliquot **50–100 mL** $CuSO_4$ per group jar
- [ ] One 9 V battery per group (fresh if possible)
- [ ] Waste bucket labeled "copper waste — do not drain"
- [ ] Demo cell ready for hook (jar + electrodes + meter)

Open questions / notes

- $CuSO_4$ concentration mixed (target 0.5 M):
- Resistor value used in pilot:
- Best demo object (nail vs. key):